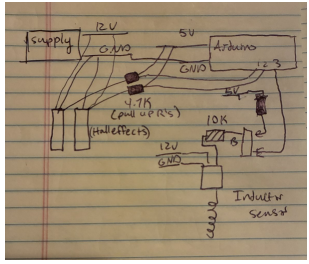
Inductive front end: attenuate vs. regenerate

What failed

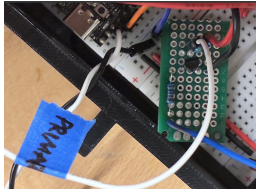
- Treated a **12 V open-collector** output as a level problem ⇒ divider + Schottky clamps.
- **147 kΩ, 82 µA** — light load, assumed the signal would survive it.
- Bench: logic-high **sagged to 2.4 V**, below threshold.
- **Topology, not values.** An open-collector stage never drives its line high — a divider only attenuates a signal that is being *driven*.

What replaced it

- **2N3904 as a level translator** — the 12 V OC output drives the base through 10 kΩ on **∼0.38 mA**.
- Transistor pulls a 3.3 V-pulled-up MCU pin low. The 12 V domain never touches the MCU.
- **Inverts** ⇒ capture on the *rising* edge.



4.7 k pull-ups to 5 V; 10 k into the base



Rebuilt front end on perfboard

Rear CAN node — role and MCU selection

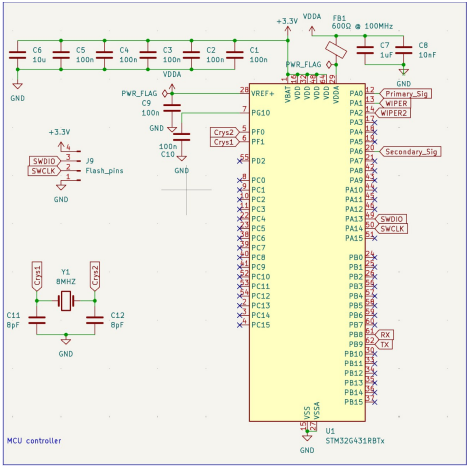
What this node does

- Acquires **engine RPM**, **CVT secondary RPM**, **two rear shock pots**; publishes to the vehicle CAN bus. One of three nodes around the central DAQ.

Why STM32G431RBTx

- **Native FDCAN** — removes an external MCP2515 and its solder joints. Fewer failure points on a chassis that vibrates.
- **Hardware timer capture** (TIM2/TIM3) timestamps edges in silicon, removing the ISR jitter the ESP32 approach carries.
- **FMAC accelerator** offloads shock-pot filtering, leaving CPU headroom for the CAN task.
- **Ecosystem** — same G4 family as the central node. One toolchain, one HAL, one debugger across every node.

Rejected: ESP32-C3 (no CAN FD), ATmega328P (needs external controller), STM32F103 "Bluepill" (no FDCAN + clone silicon risk).



MCU block: 100 nF per supply pin, 10 µF bulk, 8 MHz crystal, SWD header, ferrite-isolated VDDA

Rear CAN node — sensor front ends and analog integrity

Both open-collector — different voltage domains

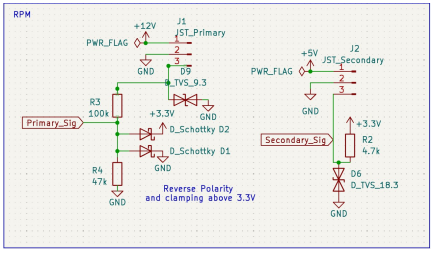
- **Primary (spark pickup, 12 V):** open-collector, one pulse per ignition event, referenced to **12 V** — ∼4× the MCU maximum. 100 k/47 k divider, Schottky clamping to the rails, **9.3 V TVS**.
- **Secondary (Hall, 5 V):** same output topology, easier domain. One **4.7 kΩ pull-up to 3.3 V** is the whole conditioning — the pull-up's rail sets the logic-high level, so no translation stage. **18.3 V TVS** to ground.

Shock potentiometers stay analog

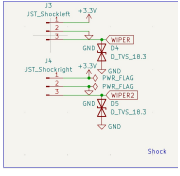
- Ratiometric divider driven by the supply ⇒ the reading *tracks the rail*. Analog-rail cleanliness *is* measurement accuracy.

Analog supply integrity

- **600 Ω@100 MHz ferrite** isolating VDDA — chosen over an inductor, which would resonate with the decoupling network.
- **1 µF ‖ 10 nF** behind it: bulk plus low-ESL.
- 100 nF per supply pin; VREF+ gets its own — reference noise lands on every conversion.



Primary vs. secondary RPM conditioning



Shock potentiometer inputs

Rear CAN node — protection, layout, open items

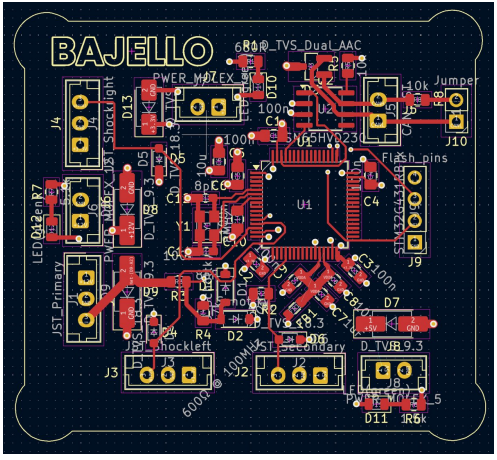
Every external pin is a way in

- TVS across **all power rails**, **both RPM inputs**, the **CAN pair** — every wire out is an antenna running past an ignition system.
- **Jumper-selectable 120 Ω termination** — which node sits at a physical bus end changes with how the car is wired.
- **LED per rail.** Turns "the node is dead" into "the 5 V rail is dead" without a multimeter in the paddock.

Layout: decoupling **placement** is the requirement, not the schematic. Deutsch DT is wire-to-wire only ⇒ Molex Micro-Fit 3.0 with pegs on the board, DT on the harness.

Open items — need a decision

- **Transceiver:** board shows SN65HVD230, ordering sheet has TJA1051T/3.
- **Inductive channel:** board keeps the *divider*; bench uses the *transistor*. Re-test first.
- Schematic through ERC, layout routed, **not fabricated**.



Layout with power and ground planes hidden — the view that shows whether signal routing is sane

Speaker script — Slide 1: inductive front end

∼60 seconds. Lead with the failure. This is the slide reviewers will ask about.

"This is the part of the design I got wrong the first time, so I want to walk through it directly.
The inductive pickup clamps onto the ignition lead and gives one open-collector pulse per spark, referenced to its 12-volt supply — roughly four times the microcontroller's absolute maximum. I read that as a level problem and reached for a divider: 147 kilohms total, drawing only 82 microamps, on the assumption that a light load would leave the signal intact.
On the bench the logic-high sagged to 2.4 volts, below threshold.
The mistake wasn't the resistor values — it was the topology. An open-collector output has no internal path to pull its line *high*; it only sinks. A divider attenuates a signal that is actively being driven, and this one never was. Different resistor values would not have fixed it.
The right circuit uses the sensor as a switch and translates domains. A 2N3904: the sensor's 12-volt open-collector output drives the base through 10 kilohms on about 0.38 milliamps, and the transistor pulls a 3.3-volt-pulled-up MCU pin low. The 12-volt domain never touches the microcontroller.
One thing to flag: the transistor inverts, so timer capture has to trigger on the rising edge — the sensor *releasing*, not firing. Easy to get backwards in firmware.
The general lesson: read the output *topology* before choosing a conditioning circuit. Open-collector, push-pull and passive outputs each fail differently, and a divider is only correct for one of them."

Speaker script — Slide 2: role and MCU selection

∼50 seconds. This is the "why this part" slide — reviewers want the reasoning, not the datasheet.

"The rear node acquires engine RPM, CVT secondary RPM, and the two rear shock potentiometers, then publishes them onto the vehicle bus. It's one of three nodes around the central DAQ.
I selected the STM32G431RBTx against our weighted decision matrix, and there were four reasons.
Native FDCAN was the big one. It removes an external MCP2515 and every solder joint that comes with it, and on a chassis that vibrates the joint count is a real reliability term.
Second, hardware timer capture. TIM2 and TIM3 timestamp sensor edges in silicon, so we don't carry the interrupt-latency jitter that the software approach on the dyno ESP32 has.
Third, the FMAC filter accelerator offloads shock-pot conditioning from the CPU, which leaves headroom for the CAN task.
Fourth — and this one is about the team, not the chip — the central node is the same G4 family. One toolchain, one HAL, one debugger across every node, so anyone who can work on one node can work on all of them.
On what I rejected: the ESP32-C3 has no CAN FD and would fragment the toolchain. The ATmega328P needs an external CAN controller. And the STM32F103 Bluepill has no FDCAN, plus the widely-sold clones carry counterfeit-silicon risk that our 35-percent durability weighting can't accept."

Speaker script — Slide 3: front ends and analog integrity

∼60 seconds. The point to land: signal type, not sensor name, determines the circuit.

"Both RPM channels are open-collector — both sink current, neither sources it. What separates them is the voltage domain they're referenced to, and that's what decides the circuit.
The primary is the spark pickup. It clamps the ignition lead and gives one open-collector pulse per ignition event, but it's referenced to its 12-volt supply — about four times the MCU's absolute maximum. So it can't touch the pin directly no matter how clean the pulse is. On this board that's a 100k/47k divider, Schottky diodes clamping to the rails, and a 9.3-volt TVS at the connector.
The secondary is the Hall sensor: same output topology, much easier domain. It runs on 5 volts and one 4.7k pull-up is the entire signal conditioning. The detail worth noting is that the pull-up's rail — not the sensor's supply — sets the logic-high level the MCU sees, so pulling it to 3.3 volts means no translation stage at all. Plus an 18.3-volt TVS.
The shock pots are the third case: they stay analog all the way to the ADC. A potentiometer is a ratiometric divider, so its reading tracks the supply rail directly — which means the cleanliness of the analog rail *is* the accuracy of the measurement.
That's why the analog supply gets its own treatment. A 600-ohm-at-100-megahertz ferrite isolates VDDA. I chose a ferrite over an inductor specifically because an inductor would form a resonant tank with the decoupling caps and ring. Behind it, 1 microfarad in parallel with 10 nanofarads — the big one for bulk load steps, the small one because it has lower ESL and still works where the big one has gone inductive. Then 100 nanofarads per supply pin, and VREF+ gets its own, because noise on the ADC reference shows up on every single conversion."

Speaker script — Slide 4: protection, layout, open items

∼60 seconds. End on the open items — that is what you want the room to respond to.

"The theme on protection is that every wire leaving this board is a long antenna running past an ignition system, and every connector is a path for a transient the MCU won't survive. So there's TVS across all power rails, both RPM inputs, and the CAN pair.

Bus termination is on a jumper rather than a fixed resistor, because which node sits at a physical end of the bus changes with how the car gets wired that season. And there's an LED on every rail — cheap, and it turns 'the node is dead' into 'the 5-volt rail is dead' without unpacking a multimeter in the paddock.

On layout: the image here has the power and ground planes hidden, which is the view that actually shows whether the routing is sane — with the pours on, everything looks connected. The decoupling requirement is placement, not schematic; each cap has to sit within a few millimetres of its pin or the bond-wire inductance it exists to cancel just wins. And connectors are a pigtail — Deutsch DT has no PCB-mount variant, so the board carries Molex Micro-Fit 3.0 with mounting pegs and the DT lives on the harness, with the pegs putting vibration load into the board instead of the solder joints.

Status: schematic complete through ERC, layout routed, not fabricated. I have two open items I want input on before it is.

First, the board currently shows an SN65HVD230 transceiver, but the ordering sheet has the TJA1051T/3 marked as ordered. Similar footprint, but the slope-control pin on one is a mode input on the other. That needs reconciling before we commit to a fab run.

Second, the board still uses the divider on the inductive channel while the bench version uses the transistor. Given that the divider is the wrong topology for an open-collector output — there's nothing driving the line high for it to attenuate — I'd like that re-tested against the actual sensor before this revision goes out."